LESSON 10.5

Math is Boundless

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can create and describe geometric designs using repetition, patterns, and rhythm, and use a pattern unit to predict what comes next.

WHY THIS MATTERS Geometry is an integral part of design. Geometric patterns have been used to decorate homes and buildings for centuries — in art, architecture, and clothing. This lesson looks at three elements of geometric design: repetition, pattern, and rhythm.

REMEMBER

Math is Boundless

Tessellations, Geometry & Mathematical Art The first element of design is repetition: the same object or shape repeated multiple times. Because it involves one object, it has predictability. The second element is a pattern: a set of objects or shapes repeated and arranged in a regular order.

1. Tessellation: Repeating pattern of polygons covering a plane with zero gaps and no overlaps.
2. Interior angles around every vertex must sum to exactly 360°.
3. Use translations, rotations, and reflections to generate symmetrical tile patterns

Words: repetition *(repetición)* · pattern *(patrón)* · pattern unit *(unidad del patrón)* · rhythm *(ritmo)* · predictability *(previsibilidad)*

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each shape: can it tessellate by itself?

Groups: Tessellates by itself · Cannot tessellate alone

_____ Squares

_____ Equilateral triangles

_____ Regular hexagons

_____ Circles

_____ Regular pentagons

_____ Regular octagons

HINT Check the angles meeting at one corner point: they must total exactly 360°.

2 **Complete the table. Show how you found each value.**

The border pattern repeats circle, square, triangle. Name the shape at each position.

Position number	Thinking with the unit of 3	Shape
7		
12		
20		

HINT The pattern unit has 3 shapes — divide the position by 3 and look at the remainder.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A bicycle has a gear ratio of 3:1. For each turn of the pedals, the rear wheel turns…

(A) 1 time
(B) 1/3 of a turn
(C) 3 times
(D) 4 times

PER 1

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

4 **Circle the letter of the best answer. Show how you know.**

Which ratio is equivalent to 2:3?

(A) 3:2
(B) 4:5
(C) 5:6
(D) 8:12

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Two wheels turn the same number of times, but one has a larger diameter. The larger wheel…

(A) travels farther
(B) travels the same distance
(C) travels less far
(D) does not move

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Pick a card you called rhythm and your partner called repetition. What is the pattern unit, and what would have to change for it to stop being predictable?

SENTENCE STARTER The pattern unit is ___, so this is ___ rather than ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 10.5

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WHY THIS MATTERS Geometry is an integral part of design. Geometric patterns have been used to decorate homes and buildings for centuries — in art, architecture, and clothing. This lesson looks at three elements of geometric design: repetition, pattern, and rhythm.

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which statement best matches this lesson's summary of 'Math is Boundless'?

(A) Designs only count as mathematical if they are perfectly predictable

(B) Rhythm is the only real element of design

(C) Geometric patterns were invented recently

(D) Math can be used to create beautiful designs, which often have repetition, patterns, and rhythm

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

With a gear ratio of 4:1, how many rear-wheel rotations come from 5 pedal rotations?

(A) 1.25

(B) 5

(C) 9

(D) 20

PER 1

3 **Circle the letter of the best answer. Show how you know.**

An artist's painting shows rows of circles. The colors and sizes of the circles vary, as does the number of concentric circles within each circle. The painting has rhythm. In the circle painting, the colors, sizes, and ring counts change from circle to circle with no repeating unit. What design element does this describe?

(A) Repetition
(B) Rhythm
(C) A pattern unit
(D) Predictability

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

An artist's painting shows rows of circles. The colors and sizes of the circles vary, as does the number of concentric circles within each circle. The painting has rhythm. Why can you NOT predict what the next circle in the painting will look like?

(A) Because the artist used no shapes at all in the painting
(B) Because the painting contains too many pattern units to count
(C) Because circles are not allowed to be part of a geometric design
(D) Because rhythm varies color, size, and the number of rings instead of repeating a fixed unit

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

An artist's painting shows rows of circles. The colors and sizes of the circles vary, as does the number of concentric circles within each circle. The painting has rhythm. A new row of circles is being added to the painting. Which addition keeps the design consistent with rhythm rather than turning it into a pattern?

(A) New circles with different colors, sizes, and ring counts than any existing row
(B) An exact copy of the very first row, repeated precisely
(C) Circles that follow a strict, repeating 3-shape unit
(D) Replacing all the circles with a single repeated square

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Pick a card you called rhythm and your partner called repetition. What is the pattern unit, and what would have to change for it to stop being predictable?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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